
When Sharing Fails Under Massart Noise: Near-Linear Lower Bounds for Multi-Distribution Learning

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Abstract

Personalized multi-distribution learning asks for one predictor per data source while exploiting a Bayes classifier shared by all sources. Under Massart noise, separate empirical risk minimization uses roughly kd/ϵ samples for k sources and VC dimension d . The best known dimension-dependent sharing term is only $k\sqrt{d}/\epsilon$. Its authors ask whether higher-order moment matching can close the gap.

We give a near-linear answer in the high-accuracy regime. For noise bound $\eta < 1/2$, let $\gamma = 1 - 2\eta$ and $z_\eta = 2(1 - \eta)/\eta - 1$. Define

$$K_m(z) = 1 + 2 \sum_{j=1}^m T_j(z)^2,$$

$$w_m = K_m(z_\eta)^{-1},$$

where T_j is a Chebyshev polynomial. Subject to explicit feasibility and concentration conditions, every adaptive learner requires

$$\Omega\left(\frac{km\gamma w_m}{\epsilon} d^{\frac{2m}{2m+1}}\right)$$

samples. Consequently, for every fixed $\zeta > 0$ the lower bound is $\Omega_{\eta,\zeta}(kd^{1-\zeta}/\epsilon)$ at sufficiently high accuracy. If ϵ is polynomially small in d , it becomes $\Omega_\eta(kd \exp[-C_\eta \sqrt{\log d}]/\epsilon)$. Thus sharing cannot yield a polynomial improvement over separate learning in this regime.

The key is an exact external-atom moment theorem. An arcsine law supported on label-zero biases admits a moment-matching measure with the maximum possible atom $1/K_m(z)$ at a prescribed label-one bias. Matching $2m$ moments delays distinguishability until a $(2m + 1)$ -fold coordinate collision. A randomized null and a variance-sensitive

holdout then lift this single-source barrier to k adaptively sampled sources without increasing VC dimension.

1 Introduction

Learning systems often draw data from several populations, environments, or clients. Multi-distribution learning formalizes the benefit of choosing which source to sample while requiring guarantees across all sources (Blum et al., 2017; Haghtalab et al., 2022). In the realizable setting, a shared classifier lets k sources reduce their total cost from separate learning, about kd/ϵ , to nearly $(d + k)/\epsilon$. Analogous additive rates are possible in agnostic minimax formulations (Zhang et al., 2024; Peng, 2024).

Bounded label noise lies between these two regimes. Suppose every source has the same Bayes classifier f^* in a class of VC dimension d , and its pointwise label error is at most $\eta < 1/2$. A single source can be learned at the fast rate $\tilde{O}(d/[\epsilon(1 - 2\eta)])$. It is natural to ask whether the shared Bayes rule preserves the realizable benefit across sources.

Hanashiro et al. (2026) recently showed that the answer depends on the benchmark. If every personalized output must compete with its source’s unknown Bayes error, an additional $\Omega(k\sqrt{d}/\epsilon)$ term is unavoidable, even though the Bayes classifier is shared. Their lower bound draws latent label biases from two priors with the same first moment. One sample from a light coordinate has the same law under both priors, so a repeated coordinate is needed. The birthday threshold gives $\sqrt{d}$.

This leaves a wide gap from the separate-learning upper bound. The source paper explicitly suggests higher-order moment matching, but identifies two obstacles. A useful alternative must keep enough mass above $1/2$ to create a fixed excess-risk gap. It must also lift to many sources while preserving one shared classifier in a class of dimension d . A constant fraction

of easily detectable high coordinates does not reach linear dependence.

We resolve these obstacles up to a subpolynomial factor when accuracy grows with dimension. The first ingredient is a sharp truncated moment result that may be useful beyond this problem. Begin with the arcsine law on a low-noise interval and prescribe an atom outside that interval. The largest atom compatible with the first $2m$ moments is the reciprocal Christoffel kernel at the external point. This atom decreases exponentially with m , at a rate that we characterize exactly. We compensate by increasing the total mass of the light coordinates. This keeps the excess-risk gap fixed while making the high coordinates rarer.

The second ingredient turns moment matching into passive sample hardness. After Poissonization, the label counts at a coordinate have likelihood proportional to an expected degree- $s + t$ polynomial of the latent bias. Matching $2m$ moments makes all count pairs with total at most $2m$ identical. Distinguishing requires a $(2m + 1)$ -fold collision, whose threshold is $d^{2m/(2m+1)}$ (Wu and Yang, 2020; Canonne, 2026).

Finally, we use disjoint source supports but a single randomized null. A source-specific alternative changes one block, and its high-coordinate indicator remains a shared Bayes classifier. A fresh holdout converts an accurate personalized output into a test. Its variance is of order $\epsilon/(1-2\eta)$, so the holdout costs only $O(1/[\epsilon(1-2\eta)])$ samples. The resulting test forces the learner to pay the collision cost on every source under the same valid null.

Contributions. Our results are entirely information theoretic.

- We identify an exact external-atom construction for $2m$ truncated moments of the arcsine law. The feasible atom has mass $1/K_m(z)$, and no larger atom is possible for this null prior.
- We obtain a parameterized personalized Massart lower bound with dimension exponent $2m/(2m+1)$. It applies to adaptive source selection and improper, randomized outputs.
- Optimizing the moment order gives $kd \exp[-O_\eta(\sqrt{\log d})]/\epsilon$ at polynomial accuracy. For any fixed exponent loss ζ , a fixed-order consequence gives $kd^{1-\zeta}/\epsilon$.
- The noise dependence is explicit through a Chebyshev kernel. This extends the prior constant-noise lower bound and isolates how the high atom shrinks as the Massart margin changes.

2 Personalized Massart Learning

Let $P_1, \dots, P_k$ be distributions on $\mathcal{X} \times \{0, 1\}$. For a binary predictor f , write $\text{err}_i(f) = P_i(f(X) \neq Y)$. We assume there is an $f^* \in \mathcal{F}$ such that, for every i and x in the support of P_i ,

$$P_i(Y \neq f^*(X) \mid X = x) \leq \eta < \frac{1}{2}. \quad (1)$$

Hence f^* is Bayes optimal on every source. Let $\eta_i^* = \text{err}_i(f^*)$.

A personalized learner adaptively chooses source indices and receives an independent labeled sample from the chosen source. It returns arbitrary possibly randomized predictors $\hat{f}_1, \dots, \hat{f}_k$. Success means

$$\max_{i \in [k]} \{\text{err}_i(\hat{f}_i) - \eta_i^*\} \leq \epsilon \quad (2)$$

with probability at least $1 - \delta$. This is the MDL-MASS objective of Hanashiro et al. (2026). Our lower bound allows improper outputs and fully adaptive source choices.

Write $\gamma = 1 - 2\eta$. Learning each source separately by VC empirical risk minimization gives the benchmark

$$O\left(\frac{k[d \log(1/\epsilon) + \log(k/\delta)]}{\gamma \epsilon}\right). \quad (3)$$

The fast $1/\epsilon$ dependence follows from the Massart variance condition (Massart and Nédélec, 2006; Boucheron et al., 2005).

Let T_j denote the degree- j Chebyshev polynomial of the first kind. Define

$$z_\eta = \frac{2(1-\eta)}{\eta} - 1 > 1, \quad \gamma = 1 - 2\eta, \quad (4)$$

$$K_m(z) = 1 + 2 \sum_{j=1}^m T_j(z)^2, \quad w_m = K_m(z_\eta)^{-1}. \quad (5)$$

The kernel has a useful closed form. With $A_\eta = \text{arcosh}(z_\eta)$,

$$K_m(z_\eta) = m + \frac{1}{2} + \frac{\sinh((2m+1)A_\eta)}{2 \sinh(A_\eta)}. \quad (6)$$

In terms of the Massart margin, $z_\eta = (1 + 3\gamma)/(1 - \gamma)$ and $A_\eta = \sqrt{8\gamma(1 + O(\gamma))}$ as $\gamma \downarrow 0$. The exact form shows two regimes. If $m A_\eta \leq 1$, then $K_m = \Theta(m)$. If $m A_\eta \geq 1$, its growing term is of order $e^{(2m+1)A_\eta}/\sinh(A_\eta)$. Thus the construction records how many moments can be matched before the prescribed label switch becomes too rare.

Theorem 1 (Parameterized lower bound). *There are universal constants $c, C > 0$ with the following prop-*

erty. Fix $\eta \in (0, 1/2)$ and an integer $m \geq 1$. If

$$\epsilon \leq c\gamma w_m, \quad (7)$$

$$dw_m \geq C, \quad (8)$$

$$m\gamma^2 w_m d^{\frac{2m}{2m+1}} \geq C, \quad (9)$$

then there is a binary class of VC dimension exactly d for which every learner satisfying (2) with probability at least 0.9 uses

$$\Omega\left(\frac{km\gamma w_m d^{\frac{2m}{2m+1}}}{\epsilon}\right) \quad (10)$$

samples in the worst case.

The three conditions have separate roles. Equation (7) keeps the hard distribution normalized. Equation (8) concentrates the number of high coordinates. Equation (9) makes the collision barrier dominate the final validation cost. All hold for the optimized choice below when η is fixed and ϵ is polynomially small in d .

Corollary 2 (No polynomial gain from sharing). *Fix $\eta \in (0, 1/2)$ and $\zeta > 0$. There are positive constants $c_{\eta, \zeta}$ and $C_{\eta, \zeta}$ such that, whenever $\epsilon \leq c_{\eta, \zeta}$ and $d \geq C_{\eta, \zeta}$, the lower bound is*

$$\Omega_{\eta, \zeta}\left(\frac{kd^{1-\zeta}}{\epsilon}\right). \quad (11)$$

Corollary 3 (Polynomial accuracy). *Fix $\eta \in (0, 1/2)$ and $\beta > 0$. For all sufficiently large d , if $\epsilon \leq d^{-\beta}$, every personalized learner can require*

$$\Omega_{\eta, \beta}\left(\frac{kd}{\epsilon} \exp[-C_{\eta} \sqrt{\log d}]\right) \quad (12)$$

samples.

Corollary 2 follows by fixing m with $(2m+1)^{-1} \leq \zeta$. For Corollary 3, take $2m \asymp \sqrt{\log(d)/A_{\eta}}$. Equation (6) gives $K_m \leq C_{\eta} e^{2mA_{\eta}}$. The losses $2mA_{\eta}$ and $\log(d)/(2m+1)$ balance at order $\sqrt{\log d}$. Polynomial accuracy makes (7) eventually valid, and the other two left sides diverge. Comparison with (3) shows that the remaining gap is subpolynomial in d , apart from logarithms.

Table 1 displays the exact tradeoff at the canonical noise bound $\eta = 0.49$ used by Hanashiro et al. (2026). Each additional moment order raises the collision exponent but reduces the external atom, which in turn tightens the accuracy condition. These are evaluations of (5), not fitted or simulated values.

3 An Exact External Atom

Let σ be the arcsine probability measure on $[-1, 1]$,

$$d\sigma(u) = \frac{du}{\pi\sqrt{1-u^2}}. \quad (13)$$

Table 1: Moment order and hard-atom mass at $\eta = 0.49$.

m	matched	$2m/(2m+1)$	w_m
1	2	0.667	0.2994
2	4	0.800	0.1443
4	8	0.889	0.03710
8	16	0.941	0.001768

Its orthonormal polynomials are $\phi_0 = 1$ and $\phi_j = \sqrt{2}T_j$ for $j \geq 1$. Thus $K_m(z)$ in (5) is its degree- m Christoffel kernel evaluated at z .

Lemma 4 (Exact external-atom moments). *For every $z > 1$ and $m \geq 1$, there is a probability measure ν_m supported on $[-1, 1]$ such that*

$$\sigma_1 = \frac{1}{K_m(z)}\delta_z + \left(1 - \frac{1}{K_m(z)}\right)\nu_m \quad (14)$$

matches the moments of σ through degree $2m$. Moreover, $1/K_m(z)$ is the largest possible mass at z among all matching probability measures whose remaining mass is supported on $[-1, 1]$.

Proof. Set $w = 1/K_m(z)$ and define, on polynomials of degree at most $2m$,

$$L(P) = \int P d\sigma - wP(z).$$

Every $P \geq 0$ on $[-1, 1]$ has a Markov–Lukacs representation

$$P(u) = s(u)^2 + (1-u^2)t(u)^2, \quad (15)$$

where $\deg(s) \leq m$ and $\deg(t) \leq m-1$ (Wu and Yang, 2020, Section 2.5). The reproducing-kernel inequality gives

$$s(z)^2 \leq K_m(z) \int s^2 d\sigma. \quad (16)$$

Since $z > 1$, equations (15) and (16) imply

$$\begin{aligned} L(P) &= \int s^2 d\sigma - ws(z)^2 + \int (1-u^2)t^2 d\sigma \\ &\quad + w(z^2-1)t(z)^2 \geq 0. \end{aligned}$$

The truncated Riesz representation theorem gives a nonnegative representing measure on $[-1, 1]$ for L . Its mass is $L(1) = 1-w$. Normalizing proves (14).

For maximality, any matching measure with atom w' at z satisfies $w's(z)^2 \leq \int s^2 d\sigma$ for every degree- m polynomial s . Taking the reproducing kernel $s(u) = \sum_{j=0}^m \phi_j(z)\phi_j(u)$ gives $w' \leq 1/K_m(z)$. $\square$

Map $u \mapsto \eta(u+1)/2$. The null law μ_0 is supported on $[0, \eta]$, while the external point z_{η} maps to $1-\eta$.

Lemma 4 produces

$$\mu_1 = w_m \delta_{1-\eta} + (1 - w_m) \nu_m, \quad \text{supp}(\nu_m) \subseteq [0, \eta], \quad (17)$$

with moments matching through order $2m$. Every null bias favors label zero. The distinguished alternative atom favors label one, and both have Massart margin exactly γ at the closest endpoints.

4 Collisions Hide the Bayes Rule

Consider a domain $\{0, 1, \dots, d\}$. Put covariate mass $1 - \alpha$ at zero and α/d at each light coordinate. The bias at zero is fixed below $1/2$. At each light coordinate x , draw q_x independently from μ_h for $h \in \{0, 1\}$, then keep q_x fixed and draw $Y \mid X = x \sim \text{Ber}(q_x)$ on every visit. This fixed latent bias is what makes coordinate collisions informative.

Lemma 5 (Moment-collision testing). *Any test between these two random environments with both errors at most $1/5$ requires*

$$\Omega\left(\frac{m}{\alpha} d^{\frac{2m}{2m+1}}\right) \quad (18)$$

samples, provided the displayed quantity exceeds a universal constant.

Proof idea. Poissonize the sample count with mean N . At one light coordinate, the counts of labels one and zero have conditional law

$$\text{Poi}(\lambda q) \otimes \text{Poi}(\lambda(1 - q)), \quad \lambda = N\alpha/d.$$

The mixed probability of a count pair (s, t) is

$$\frac{e^{-\lambda} \lambda^{s+t}}{s!t!} \mathbb{E}[q^t (1 - q)^s]. \quad (19)$$

This expectation is a polynomial of degree $s + t$. The two laws agree for $s + t \leq 2m$. Their total count is $\text{Poi}(\lambda)$ under both priors. Product subadditivity therefore gives

$$\text{TV}(Q_0, Q_1) \leq d \mathbb{P}(\text{Poi}(\lambda) \geq 2m + 1). \quad (20)$$

For

$$N \leq \frac{2m + 1}{4e\alpha} d^{\frac{2m}{2m+1}},$$

the Poisson Chernoff bound makes (20) at most $4^{-(2m+1)} \leq 1/64$. A standard lower-tail coupling converts any fixed-budget test into a Poissonized test at constant overhead. Full constants are in the supplement. $\square$

Set

$$\alpha = \frac{16\epsilon}{\gamma w_m}. \quad (21)$$

Condition (7) ensures $\alpha \leq 1$. Under the null, zero is Bayes optimal. Under the alternative, let H count coordinates at the atom $1 - \eta$. Then $H \sim \text{Bin}(d, w_m)$. Condition (8) and a Chernoff bound make

$$\frac{dw_m}{2} \leq H \leq 2dw_m \quad (22)$$

hold with constant high probability. On this event, the high-coordinate indicator is Bayes, while zero has excess

$$\Delta_0 = \frac{\alpha\gamma H}{d} \in [8\epsilon, 32\epsilon]. \quad (23)$$

Combining (18) and (21) gives the one-source cost $\Omega(m\gamma w_m d^{2m/(2m+1)}/\epsilon)$.

5 Forcing the Cost on Every Source

Take k disjoint blocks $\mathcal{X}_1, \dots, \mathcal{X}_k$, each containing one heavy and d light points. Let $\mathcal{F}$ contain the zero function and every indicator of a subset of one block's light points. It shatters a light block. No set of $d + 1$ points is shattered. A set containing a heavy point cannot label that point one, and a light-only set spanning two blocks cannot label points in both blocks one. Hence $\text{VCdim}(\mathcal{F}) = d$.

Each source is supported on its corresponding block with the marginal used in Section 4. Under the common null, independently draw one fixed μ_0^d bias vector for every block. Every realization is a valid Massart instance with shared Bayes classifier zero.

For the source- i alternative, replace only its bias prior by μ_1^d . The Bayes classifier is the indicator of source i 's high coordinates. It belongs to $\mathcal{F}$ and equals zero on every other source support, where it is also Bayes. Thus every realized alternative still has one shared classifier.

It remains to turn a learner into the test from Lemma 5. Let $\hat{f}_i$ be an ϵ -optimal output and define on a fresh source- i sample

$$Z = \mathbf{1}\{f_0(X) \neq Y\} - \mathbf{1}\{\hat{f}_i(X) \neq Y\}, \quad (24)$$

where f_0 is zero. Under the null, $\mathbb{E}Z \in [-\epsilon, 0]$. Under an alternative satisfying (22), $\mathbb{E}Z \geq \Delta_0 - \epsilon \geq 7\epsilon$.

The Massart margin controls validation variance. For any predictor f with excess at most ϵ ,

$$P_i(f(X) \neq f^*(X)) \leq \epsilon/\gamma. \quad (25)$$

On (22), zero disagrees with Bayes on mass at most $2\alpha w_m = 32\epsilon/\gamma$. Therefore

$$\text{Var}(Z) \leq P_i(f_0(X) \neq \hat{f}_i(X)) \leq 33\epsilon/\gamma. \quad (26)$$

Bernstein's inequality estimates $\mathbb{E}Z$ to 2ϵ from $O(1/(\gamma\epsilon))$ fresh samples. Thresholding at 3ϵ separates the two priors.

Proof of Theorem 1. Let L be a sufficiently small constant times the one-source lower bound. Fix source i , and run the MDL learner with source i supplied by an unknown testing environment. Simulate all other sources from independent null draws. If the learner asks for $L/2$ samples from source i , truncate and declare the alternative. If it finishes first, run the holdout test above.

Suppose the learner’s expected source- i count under the common null were $o(L)$. Markov’s inequality makes truncation rare under the null. Learner success and holdout concentration then make the test accept with constant high probability. Under the alternative, truncation is already a correct decision. Without truncation, event (22), learner success, and holdout concentration make the test reject with constant high probability.

The test uses at most $L/2 + O(1/(\gamma\epsilon))$ samples from its unknown oracle. Condition (9) makes this less than the threshold in Lemma 5, a contradiction. Hence the expected count from every source is $\Omega(L)$ under the same common randomized null. Summing over i proves (10). Pointwise correctness of the learner justifies averaging over this null prior. $\square$

6 Context and Limitations

The moment-pair technique is classical. Polynomial approximation and moment duality underlie many sharp statistical lower bounds (Wu and Yang, 2019, 2020). Recent exposition makes explicit that matching r moments delays testing until an $(r+1)$ -fold collision (Canonne, 2026). Our external-atom lemma identifies the exact atom compatible with the arcsine moments, then uses it to resolve the open dimension exponent in personalized Massart learning up to a subpolynomial factor.

Most MDL results optimize one hypothesis against the worst source (Haghtalab et al., 2022; Zhang et al., 2024; Peng, 2024). Personalized realizable learning admits additive dependence on d and k (Blum et al., 2017). If different sources may have different target classifiers, separate-learning dependence can return (Deng and Qiao, 2024). Here the target is shared. The hardness comes only from adapting to each source’s unknown Bayes error. Active and limited-round MDL address complementary access restrictions (Zhang and Zhou, 2025; Haghtalab et al., 2025).

The accuracy condition is substantive. At fixed ϵ , increasing m eventually makes the external atom too rare to maintain the risk gap while keeping $\alpha \leq 1$. Our theorem then gives a fixed exponent below one, not a uniform $d^{1-o(1)}$ statement. Determining the ex-

act joint frontier for fixed accuracy remains open. The explicit kernel may help with that question.

Our lower bound is also information theoretic. It does not impose round, properness, or computational restrictions, and it does not propose a better algorithm. Its comparison point is separate ERM, which already matches the new lower bound up to a subpolynomial dimension factor in the stated regime.

References

- Blum, A., Haghtalab, N., Procaccia, A. D., and Qiao, M. (2017). Collaborative PAC learning. In *Advances in Neural Information Processing Systems*, volume 30.
- Boucheron, S., Bousquet, O., and Lugosi, G. (2005). Theory of classification: A survey of some recent advances. *ESAIM: Probability and Statistics*, 9:323–375.
- Canonne, C. L. (2026). A short note on (distribution) testing lower bounds via polynomials. Technical Report TR26-009, Electronic Colloquium on Computational Complexity.
- Deng, Y. and Qiao, M. (2024). Collaborative learning with different labeling functions. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 10530–10552.
- Haghtalab, N., Jordan, M. I., and Zhao, E. (2022). On-demand sampling: Learning optimally from multiple distributions. In *Advances in Neural Information Processing Systems*, volume 35, pages 406–419.
- Haghtalab, N., Montasser, O., and Qiao, M. (2025). Sample-adaptivity tradeoff in on-demand sampling. In *The Thirty-Ninth Annual Conference on Neural Information Processing Systems*.
- Hanashiro, R., Shetty, A., and Jaillet, P. (2026). Is multi-distribution learning as easy as PAC learning: Sharp rates with bounded label noise. In *Proceedings of the 39th Annual Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 3109–3142.
- Massart, P. and Nédélec, É. (2006). Risk bounds for statistical learning. *The Annals of Statistics*, 34(5):2326–2366.
- Peng, B. (2024). The sample complexity of multi-distribution learning. In *Proceedings of the 37th Annual Conference on Learning Theory*, volume 247 of *Proceedings of Machine Learning Research*, pages 4185–4204.
- Wu, Y. and Yang, P. (2019). Chebyshev polynomials, moment matching, and optimal estimation of the unseen. *The Annals of Statistics*, 47(2):857–883.

- Wu, Y. and Yang, P. (2020). Polynomial methods in statistical inference: Theory and practice. *Foundations and Trends in Communications and Information Theory*, 17(4):402–586.
- Zhang, C. and Zhou, Y. (2025). Towards fundamental limits for active multi-distribution learning. In *Proceedings of the 38th Annual Conference on Learning Theory*, volume 291 of *Proceedings of Machine Learning Research*, pages 6041–6090.
- Zhang, Z., Zhan, W., Chen, Y., Du, S. S., and Lee, J. D. (2024). Optimal multi-distribution learning. In *Proceedings of the 37th Annual Conference on Learning Theory*, volume 247 of *Proceedings of Machine Learning Research*, pages 5220–5223.